Ximena Aragón Toledo

Food Engineer

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PROFILE

Food Engineer with experience in production, product development and process optimization in the food industry: running production processes, monitoring critical parameters, standardizing processes, and ensuring quality and food safety. Developed the «Zumbido» mead end to end and optimized kombucha clarification through microfiltration. Focused on continuous improvement, problem solving and meeting operational goals.

EDUCATION

Universidad Tecnológica de la Mixteca — Huajuapan de León, Oaxaca, Mexico
B.Eng. in Food Engineering — *2021 – 2026*

- Hands-on pilot-plant training: more than 20 products across the meat, dairy, cereal, and fruit and vegetable lines.

PROFESSIONAL EXPERIENCE

EtnoFood — Professional Internship — Research & Development
New Product Research & Development — *July 2025 – August 2025*

- Developed fermented products (kombucha) and new products such as a traditional salsa and tofu, from formulation to finished product.
- Monitored fermentation parameters batch by batch, standardized processes for reproducibility and packaged products under quality and food-safety criteria.
- Optimized formulations through sensory evaluation with panelists and consumer acceptance analysis.

La Fermentería — Professional Internship — Production
Brewery Production — *July 2024 – September 2024*

- Ran the craft beer production process (mashing, sparging, boiling and fermentation); monitored critical parameters and operated the equipment.
- Applied Good Manufacturing Practices (GMP) and quality controls; identified and resolved deviations through sensory evaluation.
- Promoted and sold craft beer at fairs.

Universidad Tecnológica de la Mixteca — Social Service — Chemical analysis
Synthesis of Metal Sensors for Food — *June 2023 – November 2023*

- Supported the synthesis of chemical sensors to detect copper, iron and zinc, with UV-Vis and IR spectrophotometry and thin-layer and column chromatography.

PROJECTS

Zumbido: red-fruit mead — Food Processing Pilot Plant, UTM — *January 2026 – May 2026*

- End-to-end development of a mead made with honey from Oaxaca's Sierra Sur, from concept to a standardized process, packaging and labeling for sale.

Kombucha microfiltration — Bioprocess Laboratory, UTM — *October 2025 – December 2025*

- Optimized the microbiological stabilization and clarification of kombucha through microfiltration, without compromising its functional properties.

Functional drink with antioxidant activity — Bioprocess Laboratory, UTM — *October 2024 – December 2024*

- Formulated an adaptogenic drink with nanche-leaf extract, raspberry and honey; selected the most effective formulation with the DPPH assay.

SKILLS & LANGUAGES

Production & processes: production process operation and control, process standardization and optimization, continuous improvement, problem solving
Quality & food safety: Good Manufacturing Practices (GMP), quality control, quality tools, packaging and labeling
Product development: food and beverage formulation, fermentation (beer, kombucha, mead), sensory evaluation
Laboratory: UV-Vis and IR spectrophotometry, DPPH assay, thin-layer and column chromatography, microfiltration, physicochemical analysis
Software: Microsoft Excel, Microsoft Visio, SolidWorks
Languages: Spanish (native) — English (C1, advanced)